Examiner only

5. *(a)* Sulfuric acid is manufactured using the Contact Process. The equation represents one stage of the process.

$$2SO_2(g) \; + \; O_2(g) \; \rightleftharpoons \; 2SO_3(g)$$

Describe and explain **all** the stages in the process, including the one given above, from **raw materials** to the final product, **sulfuric acid**. [6 QER]

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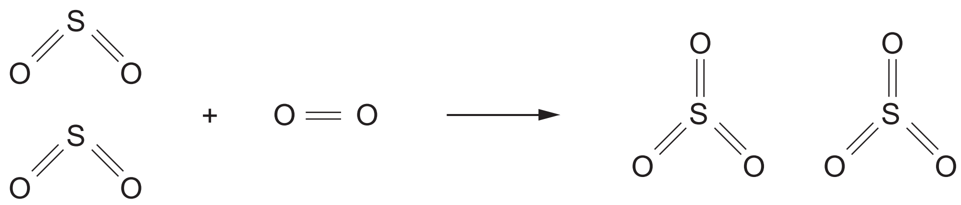
(b) The equation shows the bonds which are broken and the bonds which are formed in the production of sulfur trioxide.



The bond energy of an S ═ O bond is 523 kJ.

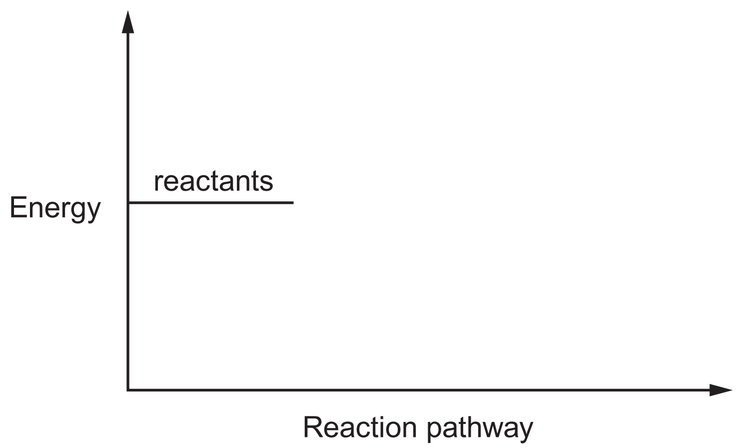
(i) The **total** energy needed to break the bonds in the reactants is 2587 kJ. Calculate the energy needed to break an O ═ O bond. [2]

Energy = ... kJ

(ii) Calculate the overall energy change for the reaction. [2]

Overall energy change =.. kJ

(iii) Complete the energy profile for the reaction. [1]



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